Introductory activity

Activity 1 · Is multiplying the same as turning? 35 min . The plane has an orthonormal frame $(O, \vec{u}, \vec{v})$. To a complex number $x + iy$ we associate the point with coordinates (x, y).

1. Plot the points A, B, C associated with $a = 2 + i$, $b = i\,a$ and $c = (1 + i)\,a$. Give b and c in algebraic form.
2. Compute the distances OA, OB and OC.
3. Compute the scalar product $\overrightarrow{OA} \cdot \overrightarrow{OB}$. What can be said about the lines (OA) and (OB)?
4. Compute $\cos\left(\overrightarrow{OA}, \overrightarrow{OC}\right)$ with the scalar product, then the sign of the determinant $\det\left(\overrightarrow{OA}, \overrightarrow{OC}\right)$. Deduce a measure of the oriented angle $\left(\overrightarrow{OA}, \overrightarrow{OC}\right)$.
5. Start again with $a = -1 + 3i$ (or a point of your choice). State a conjecture about the geometric effect of a multiplication by i, then by $1 + i$.
6. Which quantities attached to a complex number would have to be defined to state this conjecture in general?

Observing is not proving: the properties conjectured here are established in the course.